

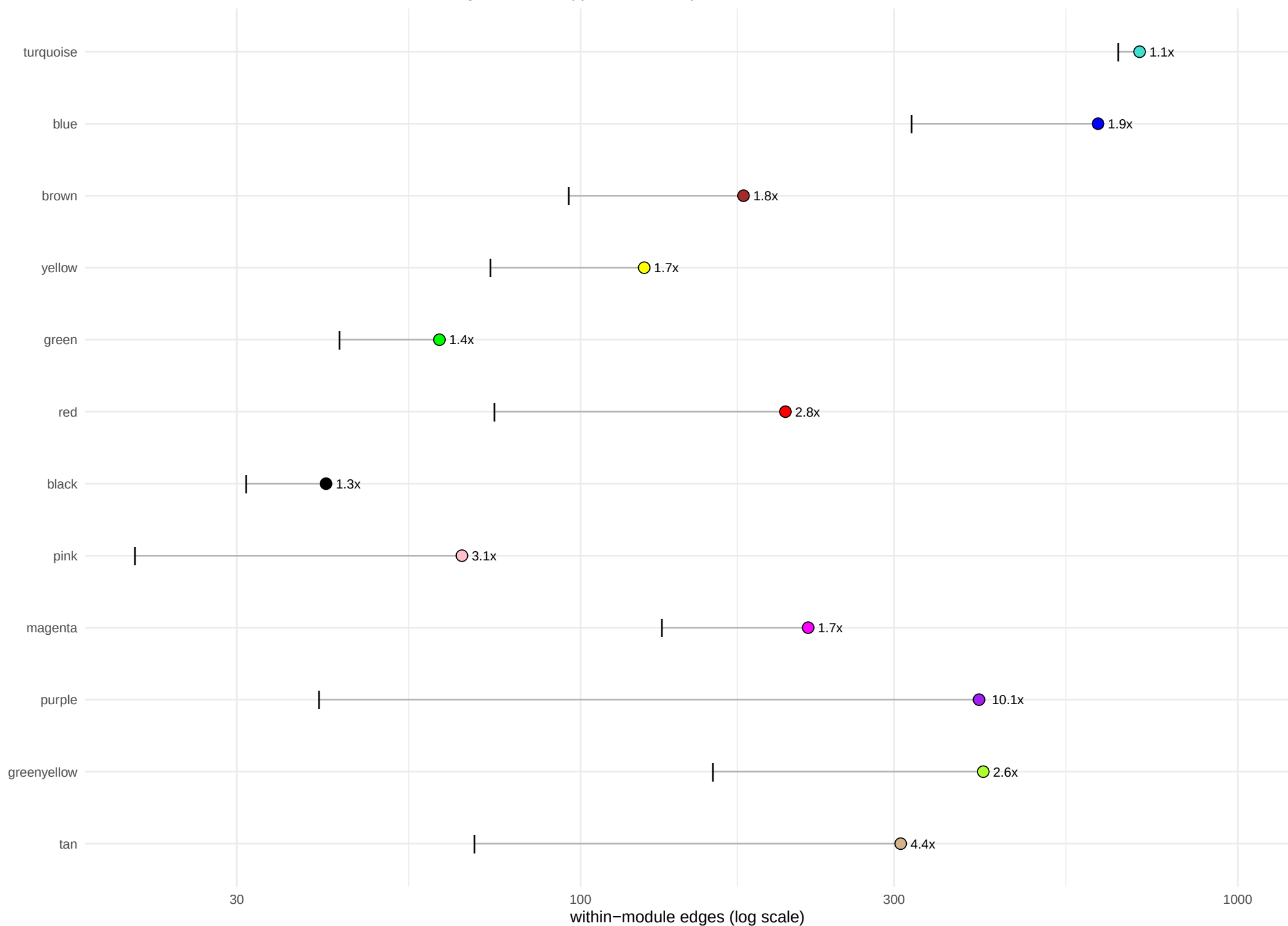
Module enrichment

clusterProfiler ORA, universe 1900 measured genes; top 3 sets per module at BH < 0.05



STRING edges within each module

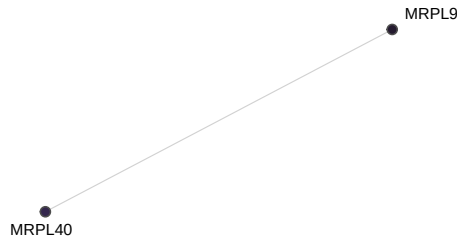
STRINGdb PPI enrichment, v12, score >= 700, background the mapped measured proteins



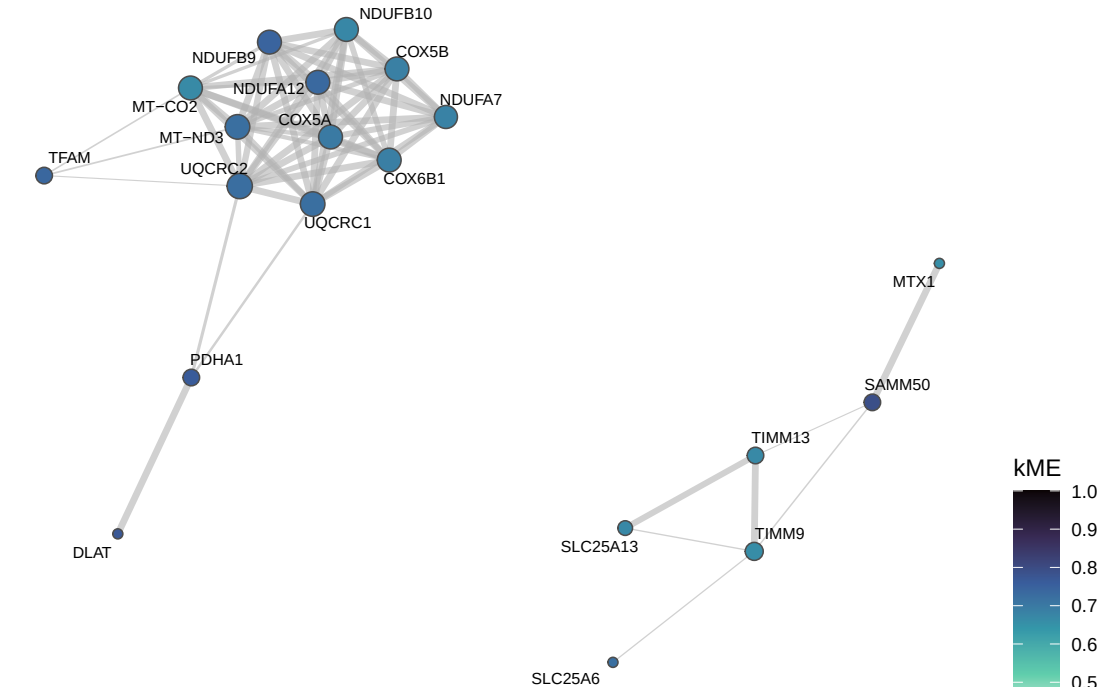
Module hub networks

top 25 members by kME per module, STRING v12 edges at score >= 700; page 1 of 3

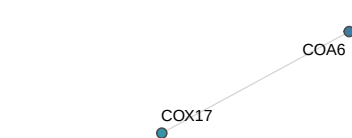
turquoise: 2 edges among 25 hubs, 21 without an edge



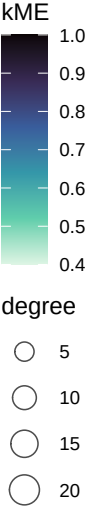
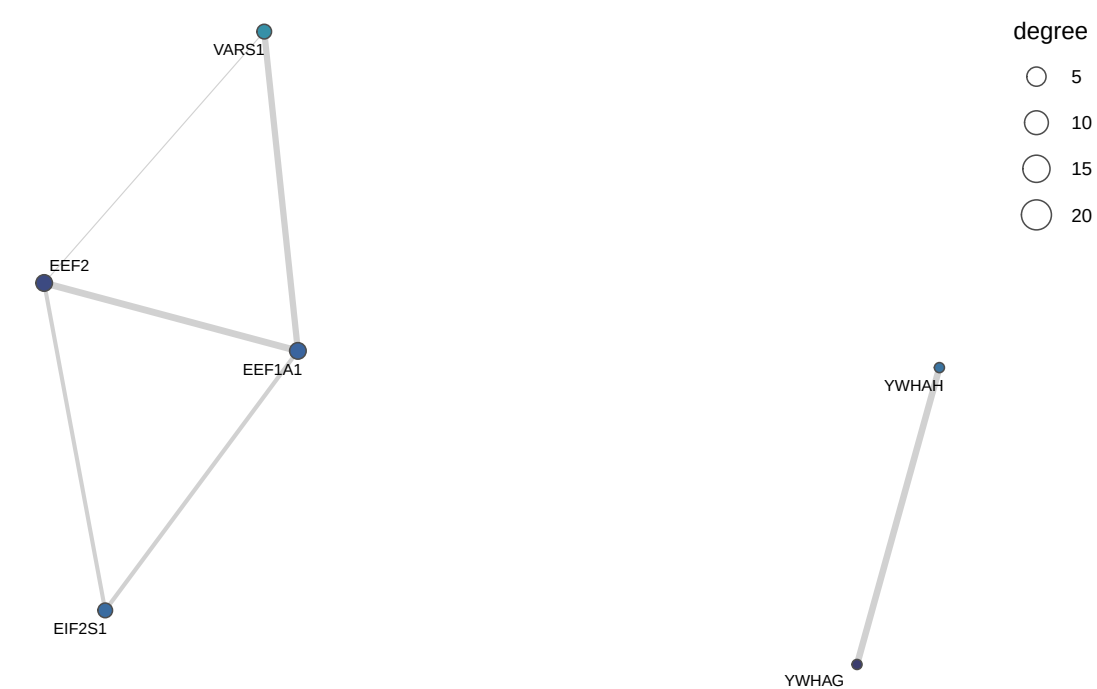
blue: 67 edges among 25 hubs, 5 without an edge



brown: 8 edges among 25 hubs, 16 without an edge



yellow: 6 edges among 25 hubs, 19 without an edge



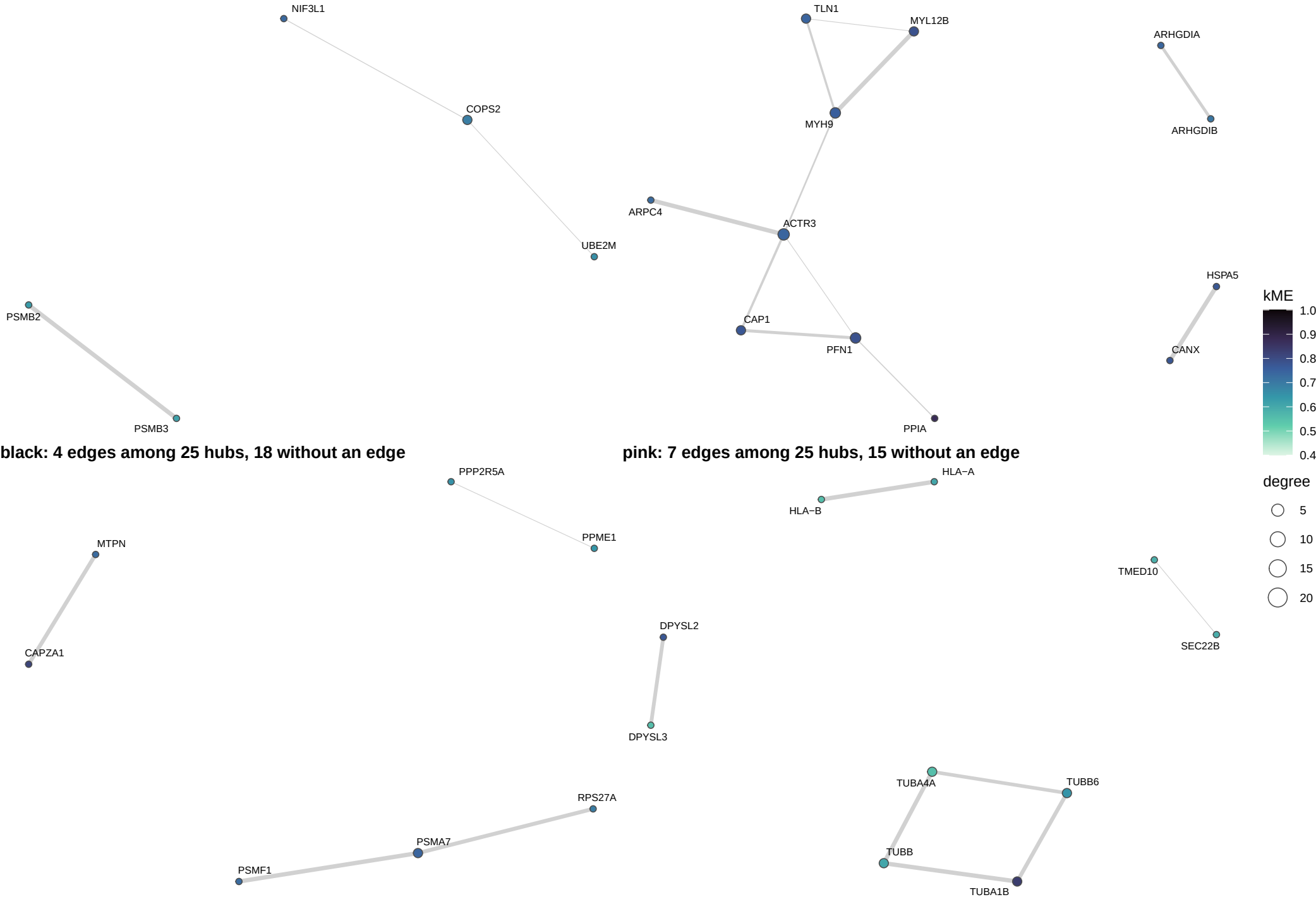
Nodes: a module's highest-kME members with at least one edge among them, fill = kME, size = degree among the drawn hubs; the title counts those without an edge. Edges: STRING combined score, width = score. Layout is force-directed and seeded; its geometry carries no meaning. Table: c_data/05_characterise.xlsx, hub_edges.

green: 3 edges among 25 hubs, 20 without an edge

red: 11 edges among 25 hubs, 13 without an edge

black: 4 edges among 25 hubs, 18 without an edge

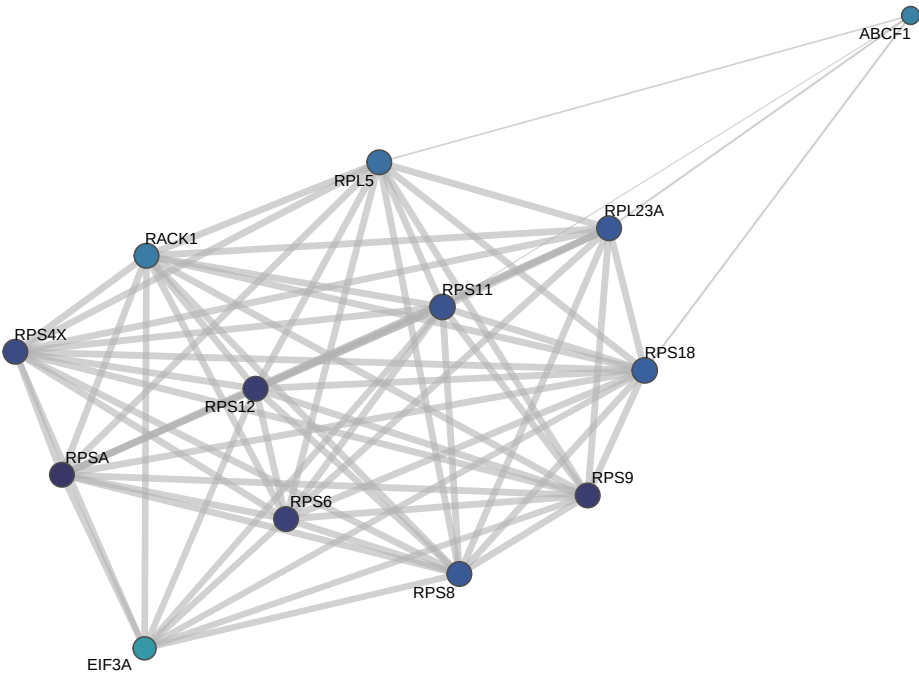
pink: 7 edges among 25 hubs, 15 without an edge



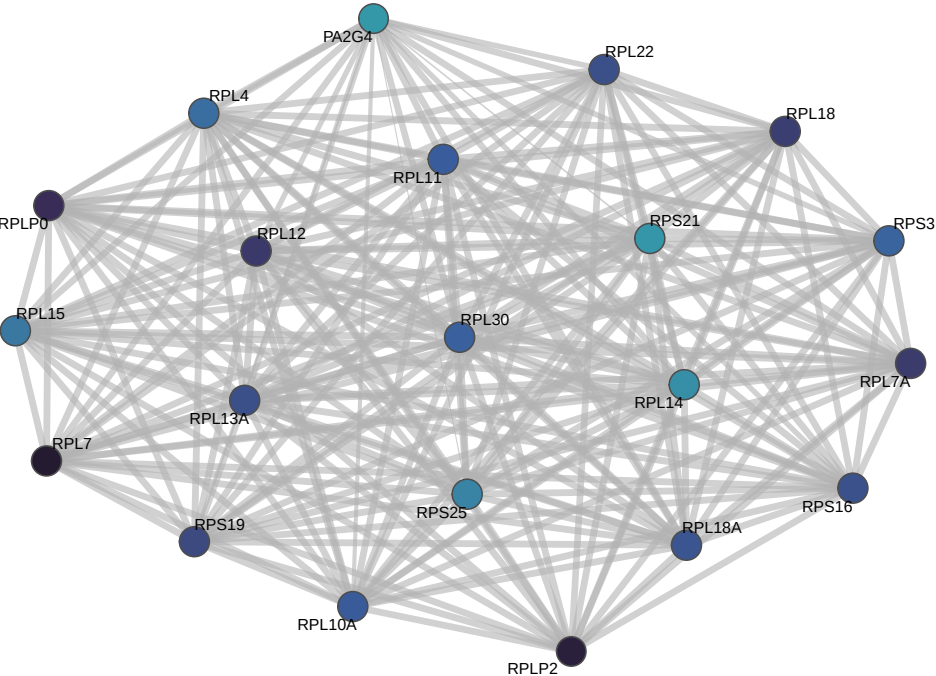
Module hub networks

top 25 members by kME per module, STRING v12 edges at score ≥ 700 ; page 3 of 3

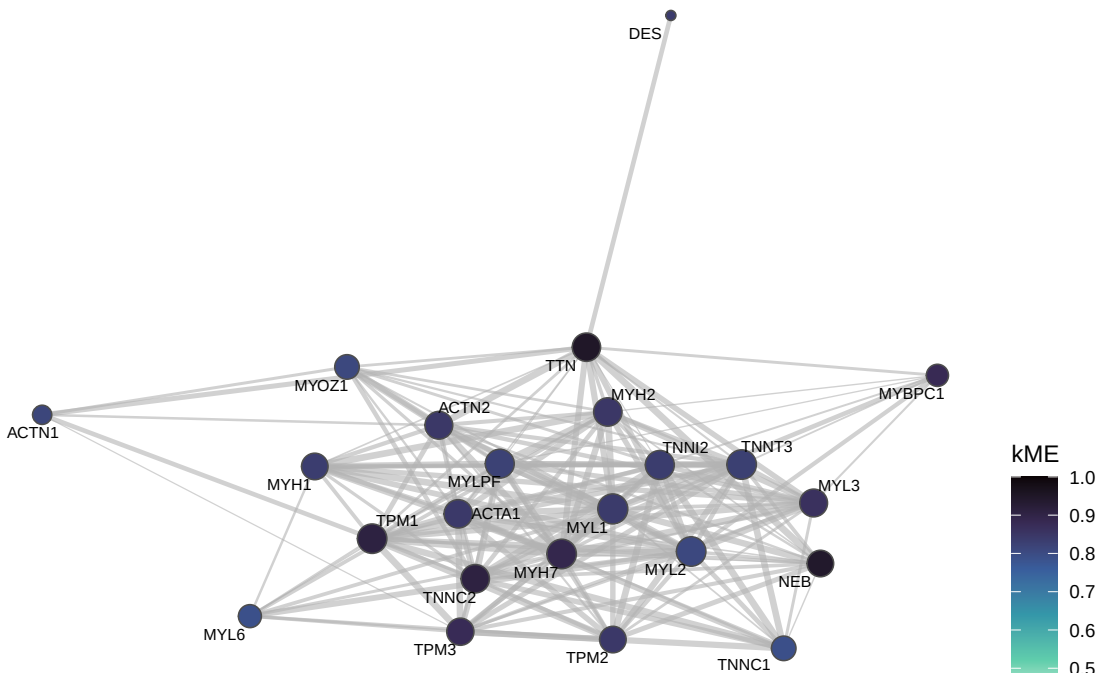
magenta: 68 edges among 25 hubs, 12 without an edge



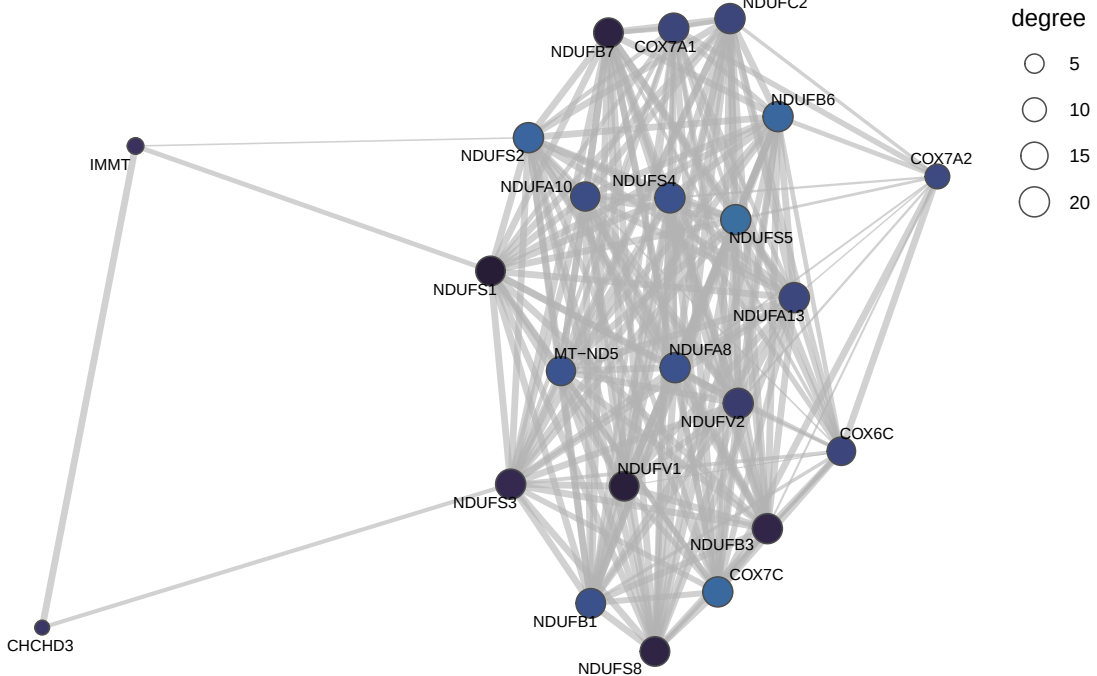
greenyellow: 209 edges among 25 hubs, 4 without an edge



purple: 167 edges among 25 hubs, 2 without an edge



tan: 202 edges among 25 hubs, 2 without an edge



Nodes: a module's highest-kME members with at least one edge among them, fill = kME, size = degree among the drawn hubs; the title counts those without an edge. Edges: STRING combined score, width = score. Layout is force-directed and seeded; its geometry carries no meaning. Table: c_data/05_characterise.xlsx, hub_edges.